

SEXUAL RISK BEHAVIOURS AND PATTERN OF MULTIPLE SITES CONCORDANCE OF HUMAN PAPILLOMAVIRUS INFECTIONS AMONG SEXUALLY ACTIVE NIGERIAN WOMEN

Imran O. Morhason-Bello^{1,2,8}, Kathy Baisley³, Isaac F. Adewole¹, Miquel Pavon⁴, Rasheed Bakare⁵,
Silvia de Sanjosé⁴ Suzanna Francis³, Deborah Watson-Jones^{6,7}

Affiliations

1. *Obstetrics and Gynaecology Department, Faculty of Clinical Sciences, College of Medicine, University of Ibadan, Ibadan, Nigeria.*
2. *Institute of Advance Medical Research and Training, College of Medicine, University of Ibadan, Ibadan, Nigeria*
3. *Department of Infectious Disease Epidemiology, Faculty of Epidemiology and Population Health, London School of Hygiene and Tropical Medicine, Keppel St, London WC1E 7HT United Kingdom*
4. *Institut Català d'Oncologia / Catalan Institute of Oncology, Cancer Epidemiology Research Program, Infections and Cancer Laboratory | Granvia de L'Hospitalet 199-203, 08908 L'Hospitalet de Llobregat, Spain | 93 2607123*
5. *Department of Microbiology, Faculty of Basic Medical Sciences, College of Medicine, University of Ibadan, Nigeria*
6. *Mwanza Intervention Trials Unit, National Institute for Medical Research, Mwanza, Tanzania*
7. *Clinical Research Department, Faculty of Infectious and Tropical Diseases, London School of Hygiene and Tropical Medicine, Keppel St, London WC1E 7HT, United Kingdom.*
8. *Presenting author*

Background

- Human papillomavirus (HPV) associated cancers are increasingly reported globally including sub-Saharan Africa (SSA); it currently accounts for 4.5% of all cancers with highest burden in low-and middle- income countries
- The persistence of HPV infection is associated with anogenital warts and in over 95.0% of cervical cancer, 88.0% of anal cancer, 70.0% of vaginal cancers, 43.0% of vulvar cancer and 25.6% of head and neck cancers
- However, data on the epidemiology of oral and other genital HPV infections with the exception of cervical HPV infection are limited in SSA

Sexual behaviour and Changing HPV burden

- Penetrative and non-penetrative sexual behaviours have been associated with HPV epidemiology
- In SSA a recent systematic review showed that adolescents and adults are increasingly engaging in unprotected vaginal, oral and anal sex

Goal

- To determine the prevalence of sexual risk behaviours, and pattern of multiple sites HPV infections among women aged 18-45 years in Ibadan, Nigeria

Specific objectives

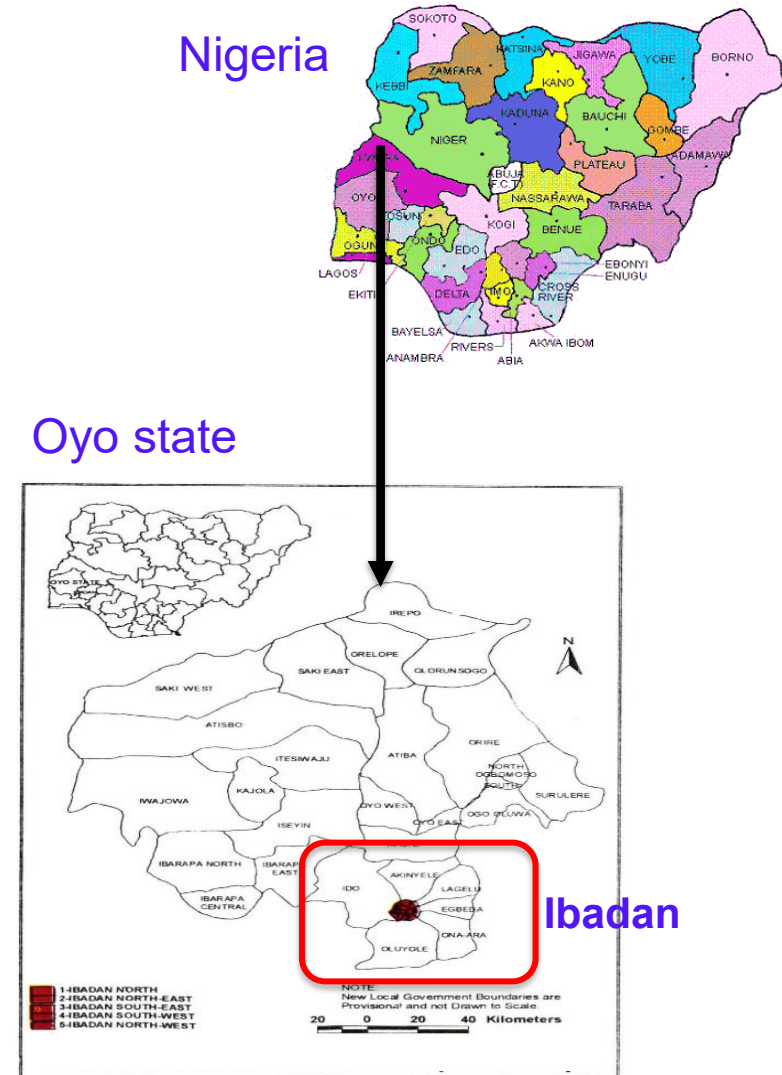
1. To determine the prevalence of sexual risk behaviours (oral, anal and unprotected penile-vaginal sex)
2. To determine the prevalence of any HPV, high-risk and low-risk HPV infections in oral, cervical, vulvar and anal samples
3. To determine factors associated with any HPV infections in each of the anatomic sites
4. To determine pattern of multiple anatomic sites concordance of HPV infections

Material and Methods

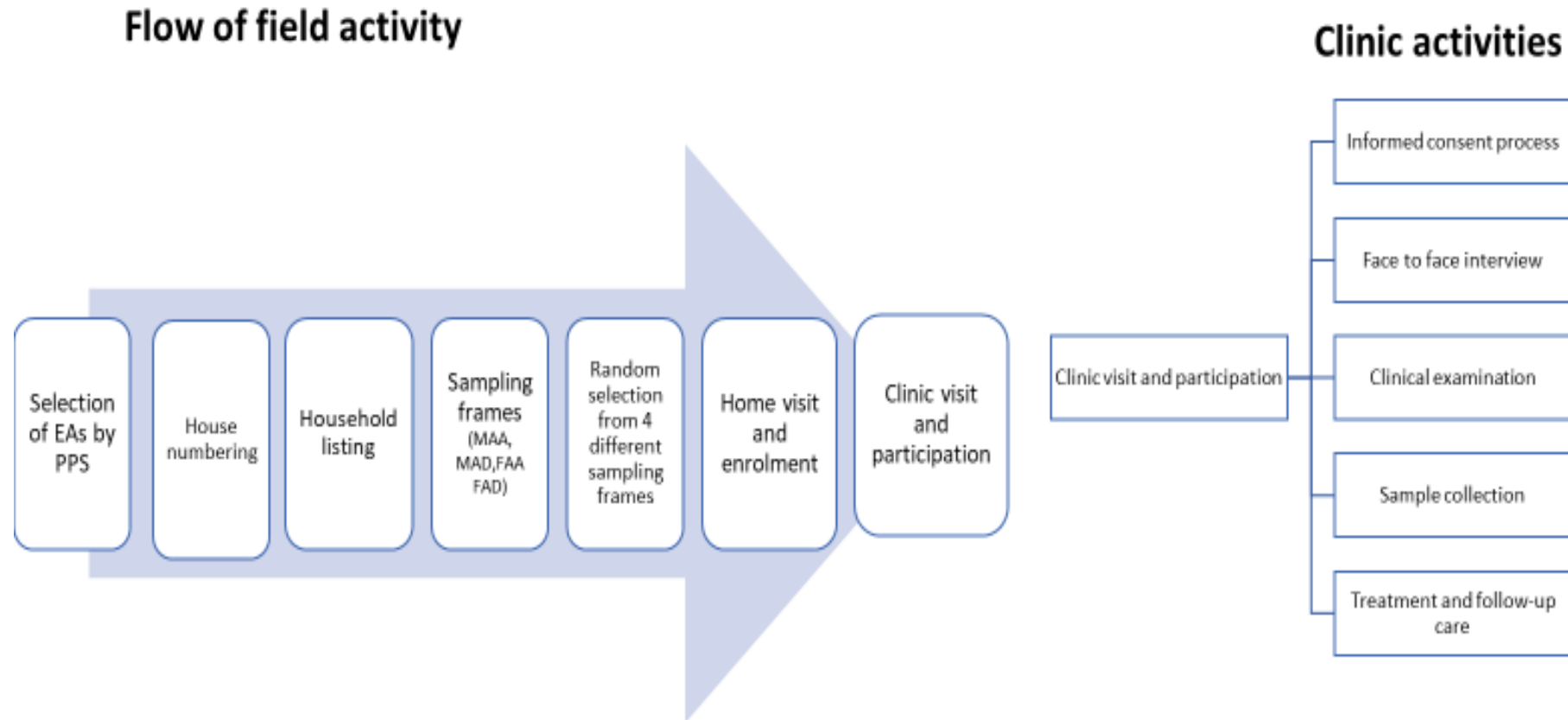
- A cross-sectional household study of hetero-sexually active women aged 18-45 years in two communities in Ibadan, Nigeria
- A face-to-face interview, clinical examination and oral, cervical, vulvar, and anal samples were collected from participants
- Anyplex II 28HPV assay was used to test samples for 28 HPV genotypes
- Descriptive and multivariable analyses were used to report prevalence and risk factors associated with HPV infection
 - Explanatory variables: Sociodemographic, Behavioural and Biological Factors
 - We used hierarchical modelling
- Analyses were performed with STATA 16.0 software

Study site

- Nigeria has a population of about 211 million people
- **Ibadan metropolis**
 - Located in SW Nigeria
 - Local Population is 2.6 million
 - Administratively divided into 11 Government Areas (LGAs)
 - Demographically, there are 6 urban LGAs while the rest are rural/peri-urban mix

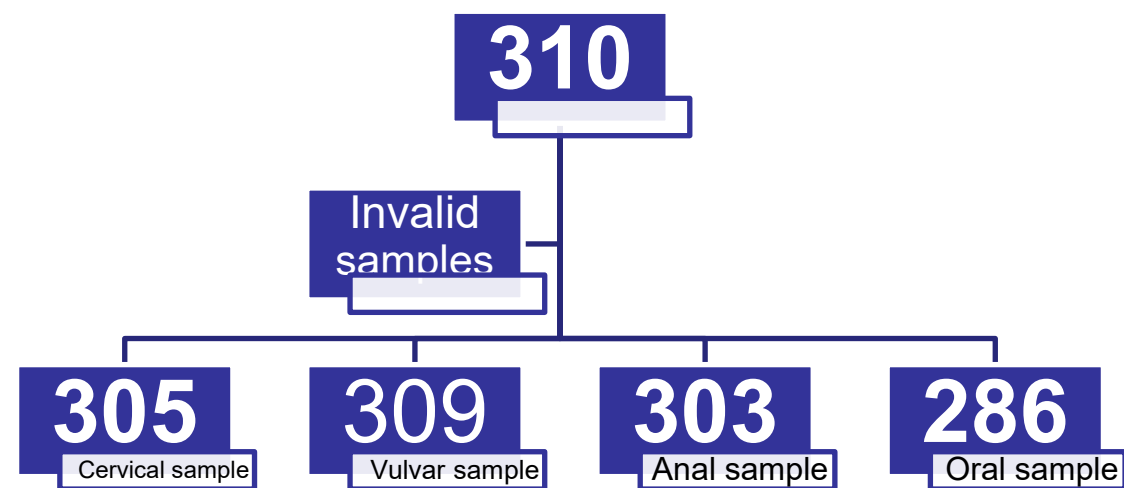


Summary of study procedure



Results

Study participants flow



Some descriptive statistics

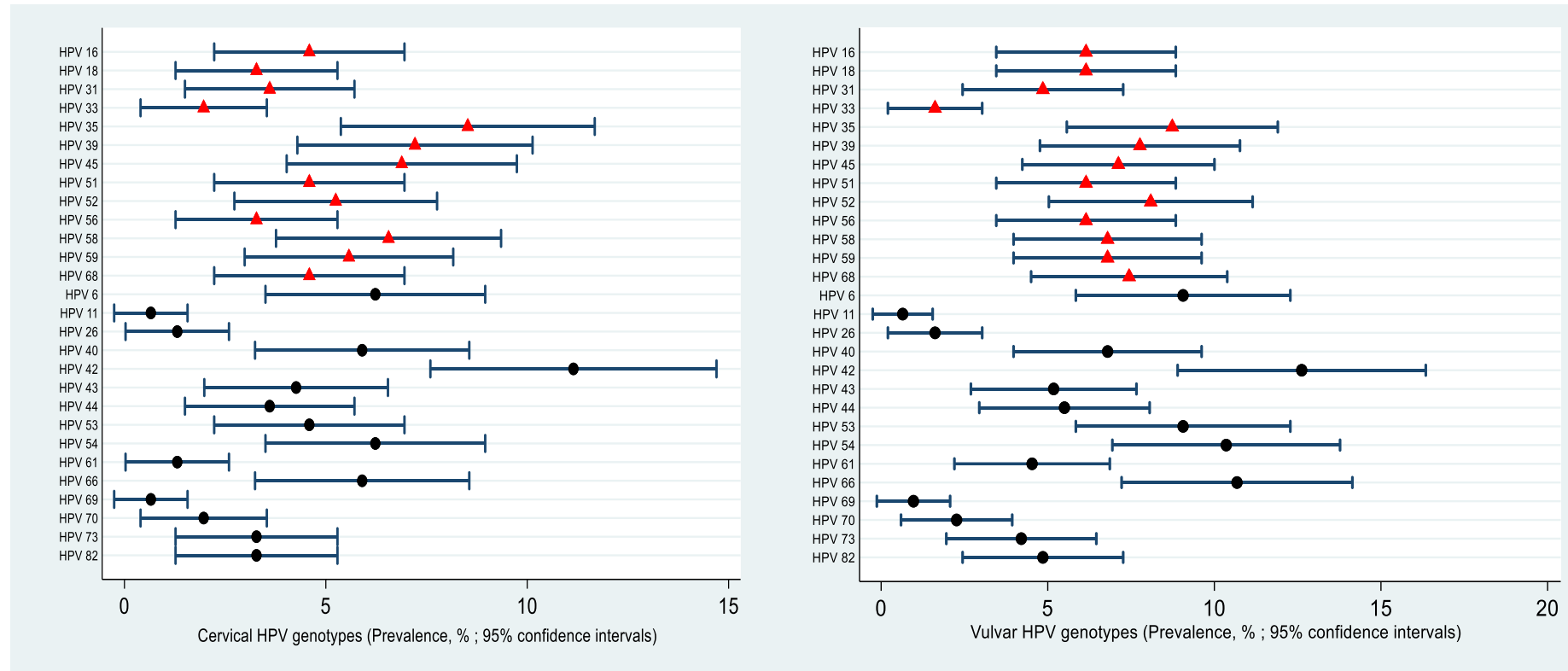
- **Mean age (SD):** 29 (7) years
- **Ethnicity:** Mostly Yoruba (240 – 77%) and others– 70 (23%)
- **Over half** had secondary education (176 – 57%)
- **A little over half earn** less than 30USD per month as income (161 – 52%)
- **About two-third** are either married or living as married (212 – 68%)

Pattern of sexual risk behaviours

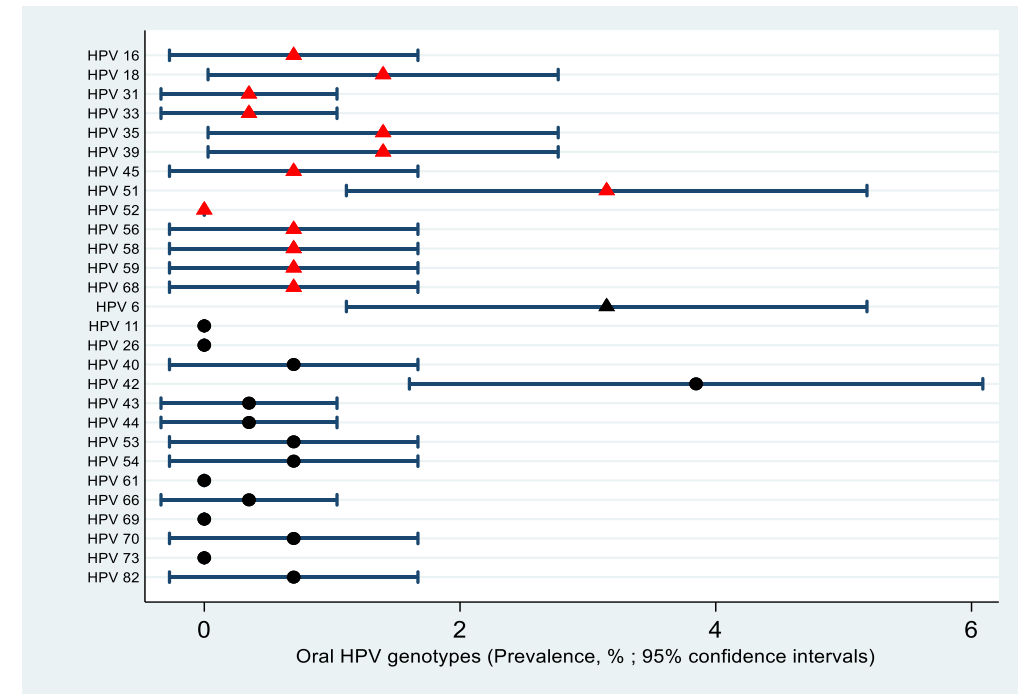
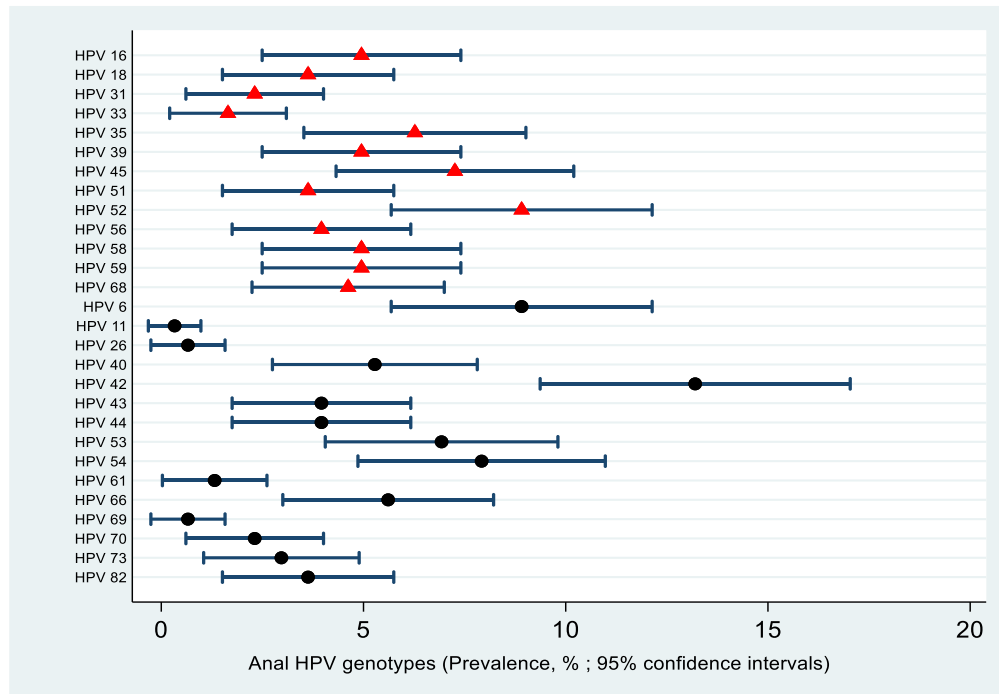
Variable	Vaginal sex	Oral sex ¹	Anal sex	Mutual Masturbation	Self Masturbation
Age at initiation, years	19 (3) ²	23.5 (6.1) ²	22 (0.0) ⁴	19((6)	
Prevalence	310 (100%)	48 (15.5%)	1 (<1.0%) ₄	234 (76%)	175 (57%)
Multiple partners	147(47%)	9 (18.8%) ³			
Average number of lifetime partners	1 (1.4) ²	2 (1.3) ^{2 3}	1 (<1%)		
Any barrier used during last sexual act	56 (18%)	3 (6.3%) ³	-		

¹ – ever gave or received oral sex; ² – Mean(SD); ³ – n=48; ⁴ – n=1; ⁵ – n=161 (age at first masturbation)

Prevalence of High-risk & Low-risk HPV by anatomic site (Cervix and Vulvar).....1



Prevalence of High-risk & Low-risk HPV by anatomic site (Anal and Oral Cavities).....2



HPV prevalence by HPV vaccine types

HPV vaccine type	Cervical Sample		Vulvar Sample		Anal Sample		Oral Sample	
	n/N	Prevalence (%) [95% CI]	n/N	Prevalence (%) [95% CI]	n/N	Prevalence (%) [95% CI]	n/N	Prevalence (%) [95% CI]
Any Bivalent HPV¹								
18-24 year	10/79	12.7 (6.2-22.0)	14/88	15.9 (9.0-25.2)	9/74	12.2 (5.7-21.8)	2/20	10.0 (1.2-31.7)
25-34 years	5/62	8.1 (2.7-17.8)	8/68	11.8 (5.2-21.9)	8/59	13.6 (6.0-25.0)	2/16	12.5 (1.6-38.3)
35-45 years	7/41	17.1 (7.2-32.1)	13/56	23.2 (13.0-36.4)	8/47	17.0 (7.6-30.8)	2/13	15.4 (1.9-45.4)
Overall	22/182	12.1 (7.7-17.7)	35/212	16.5 (11.8-22.2)	25/180	13.9 (9.2-19.8)	6/49	12.2 (4.6-24.8)
Any Quadrivalent HPV³								
18-24 year	21/79	26.6 (17.3-37.7)	26/88	29.5 (20.3-40.2)	19/74	25.7 (16.2-37.2)	6/20	30.0 (11.9-54.3)
25-34 years	8/62	12.9 (5.7-23.9)	13/68	19.1 (10.6-30.5)	13/59	22.0 (12.3-34.7)	4/16	25.0 (7.3-52.4)
35-45 years	9/41	22.0 (10.6-37.6)	18/56	32.1 (20.3-46.0)	15/47	31.9 (19.1-47.1)	5/13	38.5 (13.9-68.4)
Overall	38/182	20.9 (15.2-27.5)	57/212	26.9 (21.0-33.4)	47/180	26.1 (19.9-33.2)	15/49	30.6 (18.3-45.4)
Any Nonavalent HPV⁴								
18-24 year	45/79	57.0 (45.3-68.1)	51/88	58.0 (47.0-68.4)	44/74	59.5 (47.4-70.7)	7/20	35.0 (15.4-59.2)
25-34 years	25/62	40.3 (28.1-53.6)	32/68	47.1 (34.8-59.6)	31/59	52.5 (39.1-65.7)	6/16	37.5 (15.2-64.6)
35-45 years	18/41	43.9 (28.5-60.3)	28/56	50.0 (36.3-63.7)	23/47	48.9 (34.1-63.9)	7/13	53.8 (25.1-80.8)
Overall	88/182	48.4 (40.9-55.9)	111/212	52.4 (45.4-59.2)	98/180	54.4 (46.9-61.9)	20/49	40.8 (27.0-55.8)

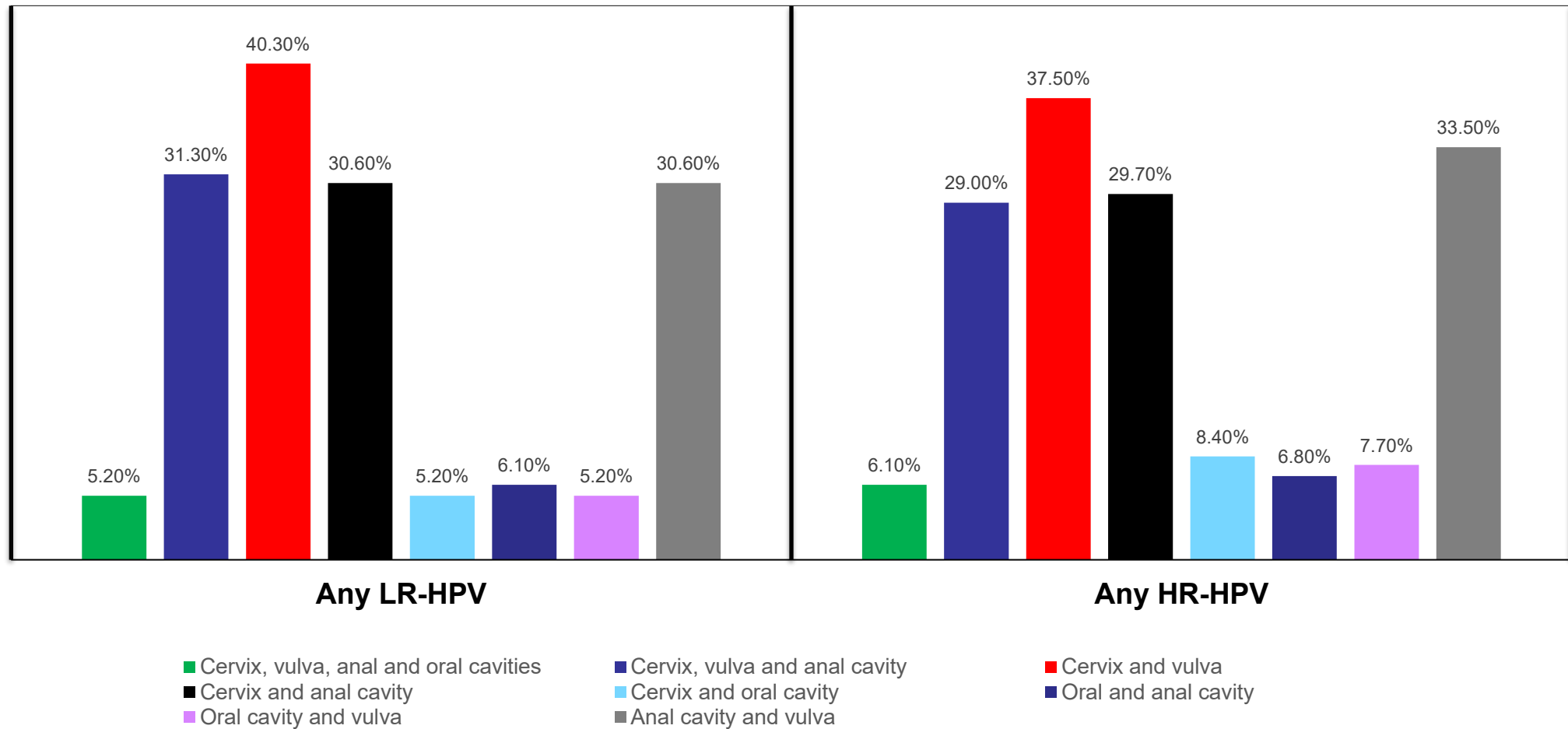
Risk Factor analyses of HPV infections by anatomic sites..1

Risk Factors	Any HPV infection (AOR (95% CI))			
	Cervical	Vulvar	Anal	Oral
Study site (Urban vs rural)	-	-	-	0.38(0.19-0.74)
Age group, years				
18-24	1	-	-	-
25-34	0.81(0.46-1.41)			
35-45	0.44(0.25-0.78)			
Age diff between first vaginal sex partner				
≤ 5	-	1	1	-
≥ 6		0.51(0.29-0.90)	0.58(0.34-1.00)	
Number of vaginal sex partners (mean (SD))	1.54(1.16-2.04)	2.03(1.38-3.01)	1.92(1.35-2.71)	-

Risk Factor analyses of HPV infections by anatomic sites..2

Risk Factors	Any HPV infection (AOR (95% CI))			
	Cervical	Vulvar	Anal	Oral
Cervical HPV infection				
No	NA	1	1	1
Yes		20.75(10.99-39.17)	4.10(1.85-9.11)	4.81(1.58-14.62)
Vulvar HPV infection				
No	1	NA	1	-
Yes	12.85(5.70-28.99)		5.85(2.35-14.59)	
Anal HPV infection				
No	1	1	NA	-
Yes	3.48(1.74-6.96)	15.31(8.12-28.86)		
Oral HPV infection				
No	1	1	-	NA
Yes	4.37(1.50-12.71)	2.53(1.13-5.67)		

Pattern of HPV concordance by anatomical sites among women in Ibadan, Nigeria (n=310)



Concluding remarks

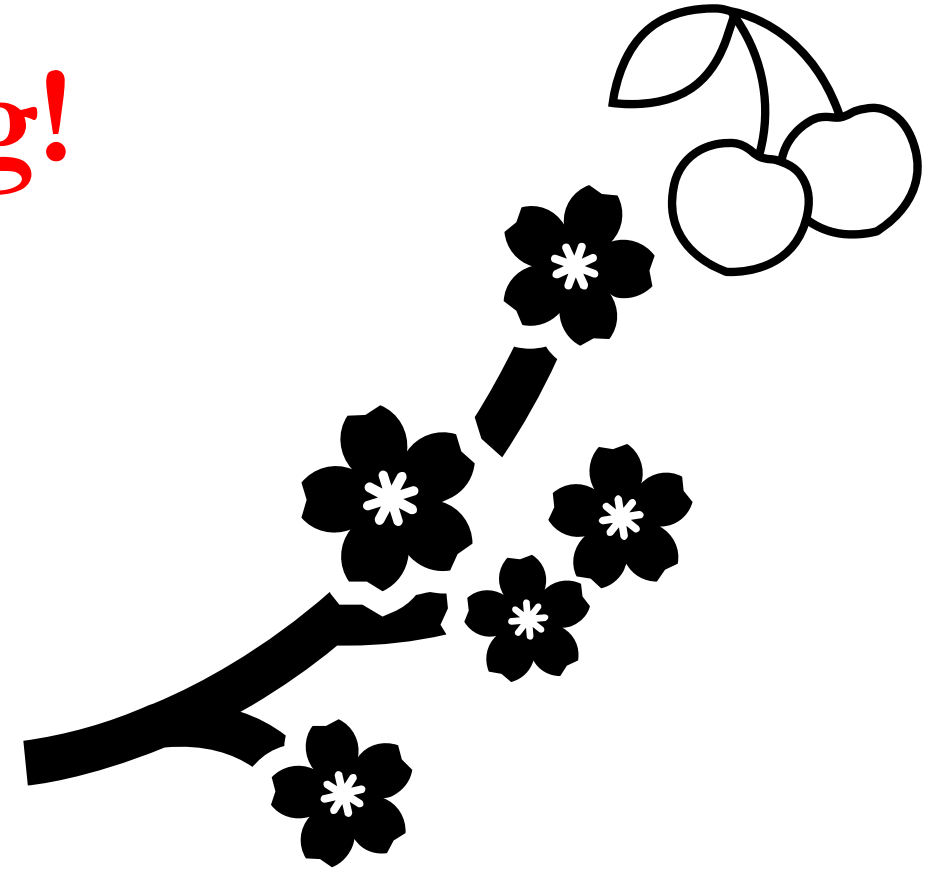
- This study shows that penetrative and non-penetrative sexual risk behaviours are common among women involved in heterosexual relationship
- The study provides a robust estimate of cervical, vulvar, anal and oral HPV prevalence among sexually active women in the general population in Nigeria and demonstrates how prevalent HPV infection is in these women
 - HPV 35 is the commonest high-risk in Cervix and Vulvar, HPV 52 in Anal samples and HPV 51 in Oral samples
- Concordance of HPV infections has programmatic and public health implication in the design of HPV associated cancers in women

Acknowledgements

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Thank you for Listening!

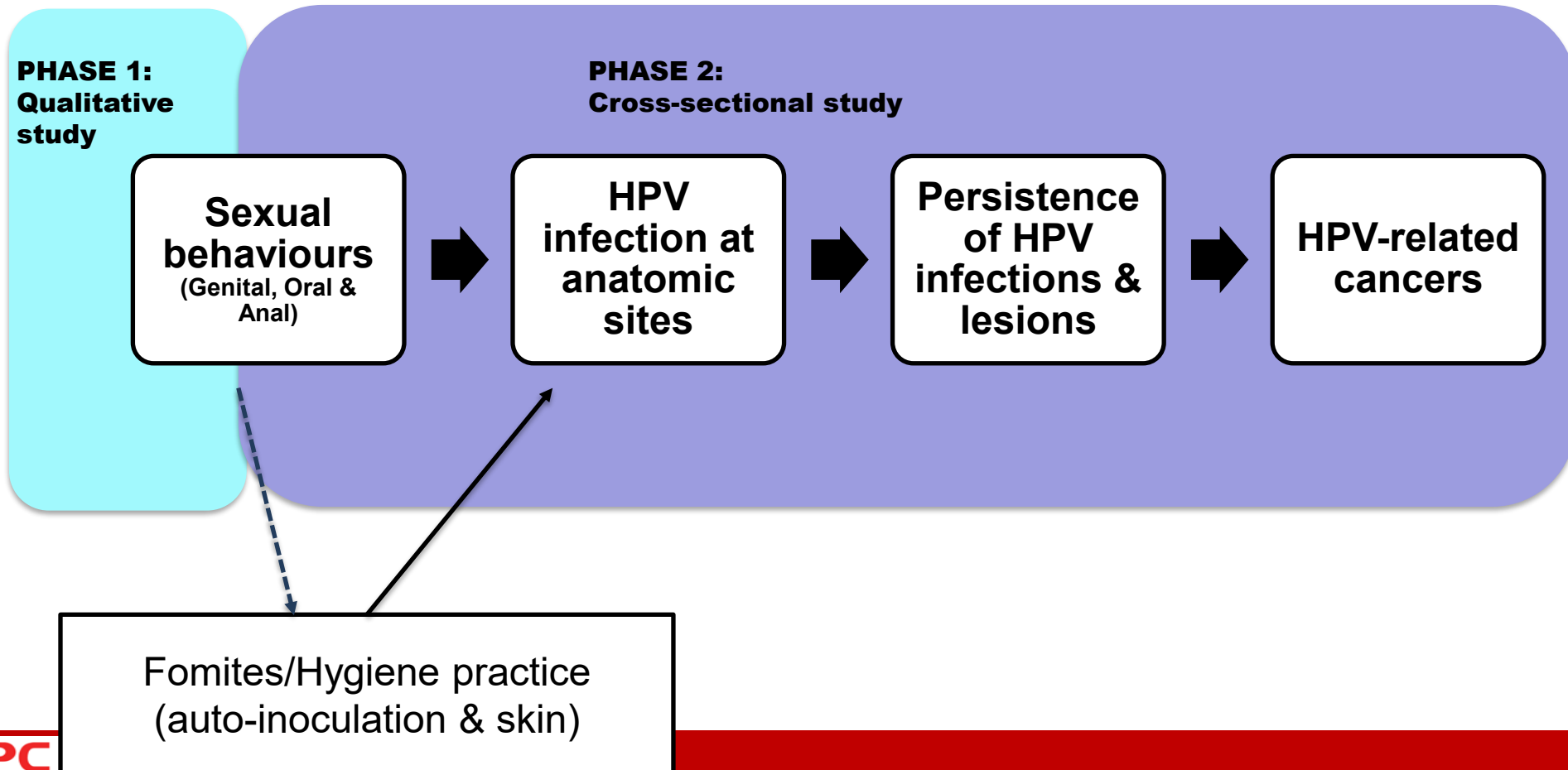


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ADDITIONAL SLIDES

To be used to answer questions

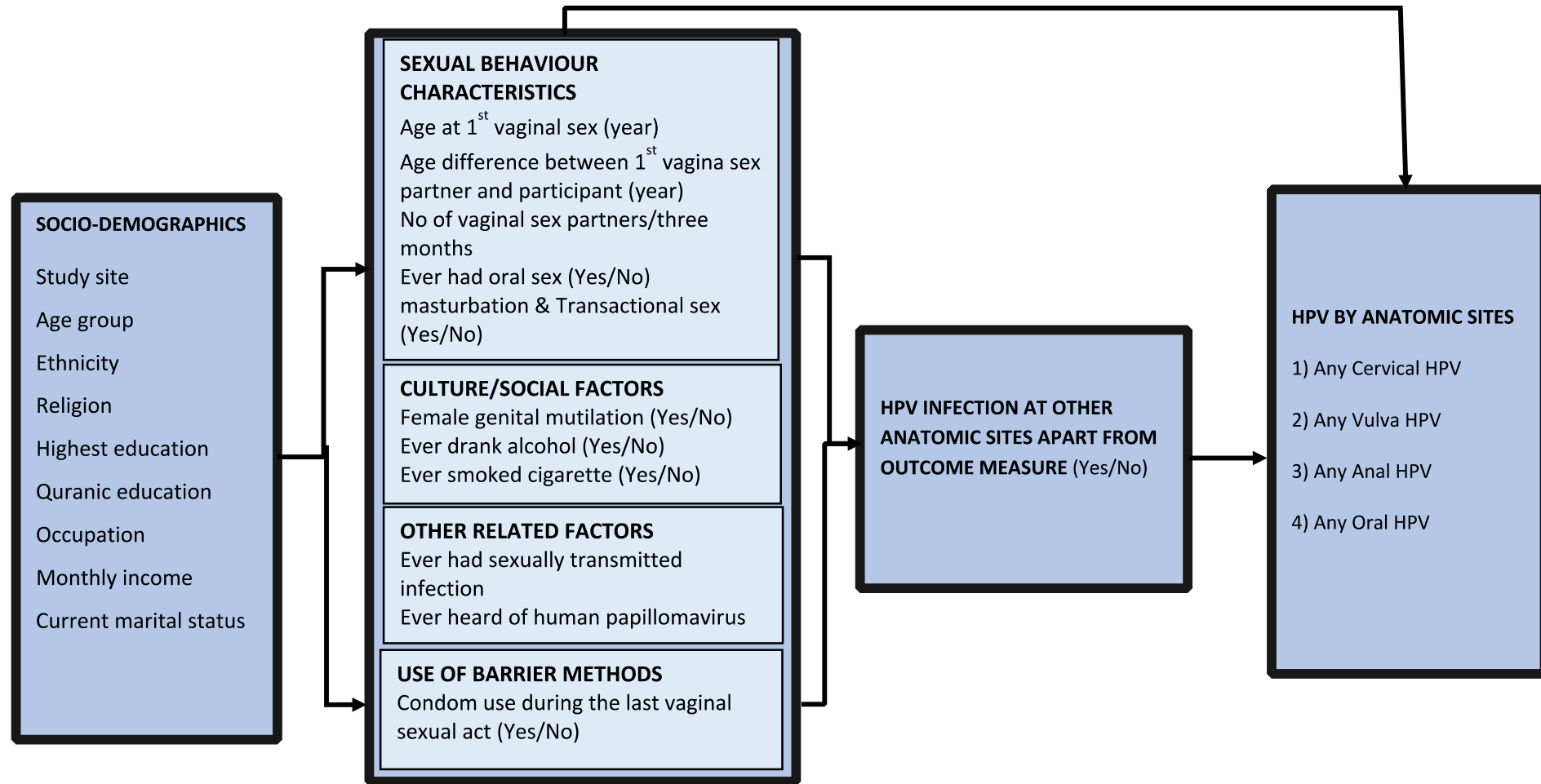
Conceptual framework for SHINI Project



Sample collection techniques

Sample	Summary procedure
Oral	30 seconds oral rinse and gargle method with 10ml of Scope mouth wash (Procter and Gamble®)
Cervical	Expose vagina and cervix disposable Cusco's speculum. Insert the tip of a wet Dacron swab into the cervical os and gently rotate 360 degrees
Vulva	Expose the vulva. Rub the tip of a wet Dacron swab on the labial and introitus on either side of the vaginal orifice without touching the urethral orifice
Anal	Insert the wet Dacron swab 4-6cm into the anal canal with participant in lateral position, and rotate it 360 degrees with gentle pressure before removing it
Blood	5 ml venous blood will be drawn into a specimen bottle

Conceptual Framework for the Data Analysis



Age Distribution of women (n=310)

Age group, years	Total n(%, column)	Mokola n(%, column)	Moniya/Sasa n(%, column)
18 - 24	121 (39%)	61 (39%)	60 (39%)
25 - 34	101 (33%)	60 (38%)	41 (27%)
35 - 45	67 (28%)	36 (23%)	52 (33%)